SOUMITRA PHARIKAL

Integrated BS-MS Degree

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EDUCATION

Year	Degree/Exam	Institute	CGPA/Marks
2029	Integrated BS-MS Degree 5Y (CGPA till now)	IISER, Berhampur	8.7 / 10
2023	Higher Secondary School Certificate	Puna International School, Gandhinagar	85%
2021	Secondary School Certificate	Delhi Public School, Ahmedabad	96.2%

INTERNSHIPS

Observational Astronomy Research Intern — Indian Centre of Space Physics — Kolkata, West Bengal **Jun 2026 - Aug 2026**

Conducted an observational astrophysics research internship under Dr. Devendra Bisht (ICSP Kolkata), focusing on the photometric study of variable stars in the Galactic open cluster IC 1369.

- Conducted photometric study of variable stars in Galactic open cluster IC 1369, reducing 814 time-series CCD frames.
- Performed DAOPHOT PSF photometry, Lomb–Scargle period analysis, and phase-folding to characterize four candidate variable stars.
- Constructed the cluster Color–Magnitude Diagram (CMD) and performed PARSEC isochrone fitting to estimate the cluster age.

Astronomy Research Trainee — University of Hawaii at Mānoa — Hawaii, US (Remote) **Jul 2025 - Aug 2025**

Participating in Intro2Astro, an 8-week research training program led by Dr. Fei Dai (UH Mānoa), focused on observational astronomy, data analysis, and exoplanetary science.

- Analyzing real datasets from TESS, Gaia, and MAST to extract transit signatures and radial velocity patterns using Python-based workflows.
- Creating scientific visualizations-scatter plots, phase-folded light curves, and time-series diagrams to interpret exoplanet orbital dynamics.
- Engaging in UNIX-based data processing, automating workflows with shell scripting, and literature reviews.

Summer Research Intern — IISER Berhampur — Berhampur, Odisha **May 2025 - Jul 2025**

Worked under Dr. Vyas Akondi (IISER Berhampur) on wavefront sensing and adaptive optics using Zernike polynomial-based analysis.

- Investigated the behavior of Zernike polynomials under pupil resizing to model optical aberrations in dynamic aperture systems.
- Implemented MATLAB-based simulations to quantify wavefront errors and optical path distortions for varying aperture geometries.
- Explored applications in ophthalmic diagnostics and microscopic imaging, analyzed the accuracy of Zernike coefficient reconstruction.

PROJECTS

Physics-Informed CT Imaging & Machine Learning for Tumour Detection **Feb 2026 - Aug 2026**

Self-initiated and completed independently; hosted on GitHub

- Developed a CT simulation pipeline using Shepp–Logan phantoms, Radon transforms, filtered back-projection, and Hounsfield Unit conversion.
- Embedded synthetic tumours and applied logistic regression using scikit-learn to classify normal and tumour-containing CT scans.
- Integrated medical imaging, inverse problem-solving, and statistical learning using Python, with findings compiled into a scientific report.

Stochastic Simulation of Probabilistic Decay Pathways Using Discrete Monte Carlo Sampling **Jun 2025 - Jul 2025**

Self-initiated and completed independently; hosted on GitHub

- Simulated radioactive decay as a stochastic process, modeling decay trajectories across initial populations $N_0 \in [10, 10^4]$.
- Analyzed deviations from the analytical exponential decay law by computing RMS fluctuations.
- Developed dynamic visualizations and statistical plots, and compiled findings into a formal report.

Reading Project — Special Theory of Relativity **May 2025 - Jul 2025**

Pursued under the mentorship of Dr. Sandeep Chatterjee, IISER Berhampur

- Explored the postulates of relativity, Lorentz transformations, and the breakdown of absolute simultaneity.
- Derived time dilation, length contraction, momentum-energy laws, and using four-vectors in Minkowski spacetime.
- Gained foundational insight into spacetime geometry, invariant intervals, and introductory tensorial notation relevant to relativistic mechanics.

AWARDS AND ACHIEVEMENTS

- Ranked 1st on the institute-wide NGPE 2026 merit list at IISER Berhampur, conducted by IAPT for undergraduate physics students.
- Qualified for INAO 2022 (Stage 2 of IOAA selection), placing among the top 256 students nationwide in IOQA 2021–22, conducted by IAPT.
- Qualified PRMO 2019–20 (Stage 1 of IMO selection), securing a top 300 rank across Classes 8–12, conducted by MTA(I).

SKILLS AND EXPERTISE

Tools: MATLAB, Maple, Bash, Jupyter Notebook, Google Colab, Overleaf, LaTeX, Git, GitHub

Languages & Libraries: Python, C, C++, Rust, Visual Basic, NumPy, SciPy, SymPy, Astropy, Pandas, Matplotlib, Seaborn, VPython, Plotly, PyDy

COURSEWORK INFORMATION

University Courses: Classical Mechanics, Quantum Physics, Math Methods in Physics-1, Electrodynamics, Calculus-1, 2 & 3, Linear Algebra

MOOCs: Understanding Einstein:The Special Theory of Relativity (Stanford Online), Python for Everybody Specialization (University of Michigan), Coding for Everyone: C and C++ Specialization; *ongoing* (UC Santa Cruz), Machine Learning Specialization; *ongoing* (Stanford Online)

POSITIONS OF RESPONSIBILITY

Club Coordinator — Pixophiles **Jan 2025 - Present**

- Serving as Coordinator of Pixophiles, the Photography Club of IISER Berhampur-overseeing all the club operations.

Club Core Member — Naxatra **Apr 2025 - Present**

- Serving as a Core Member of Naxatra, the Astronomy Club of IISER Berhampur-coordinating telescope sessions and supporting the club's educational outreach in undergraduate astronomy and astrophysics.